8. Answer: C

For a satellite in circular orbit, the loss in its total energy is of the same magnitude as the gain in its kinetic energy.

Loss in total energy

$$= \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$= \frac{1}{2}m\left(v^2 - u^2\right)$$

$$= \frac{1}{2}\left(6.9 \times 10^2\right)\left(7900^2 - 7500^2\right)$$

$$= 2.125 \times 10^9$$

$$\approx 2.1 \times 10^9 \text{ J}$$